```
C:\Windows\system32>python --version
Python 3.14.6

C:\Windows\system32>python
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print("HELLO")
HELLO
>>> exit()
```

EXP 4 NO

EXPERIMENT - 4

PACKET SNIFFING USING RAW SOCKETS IN PYTHON

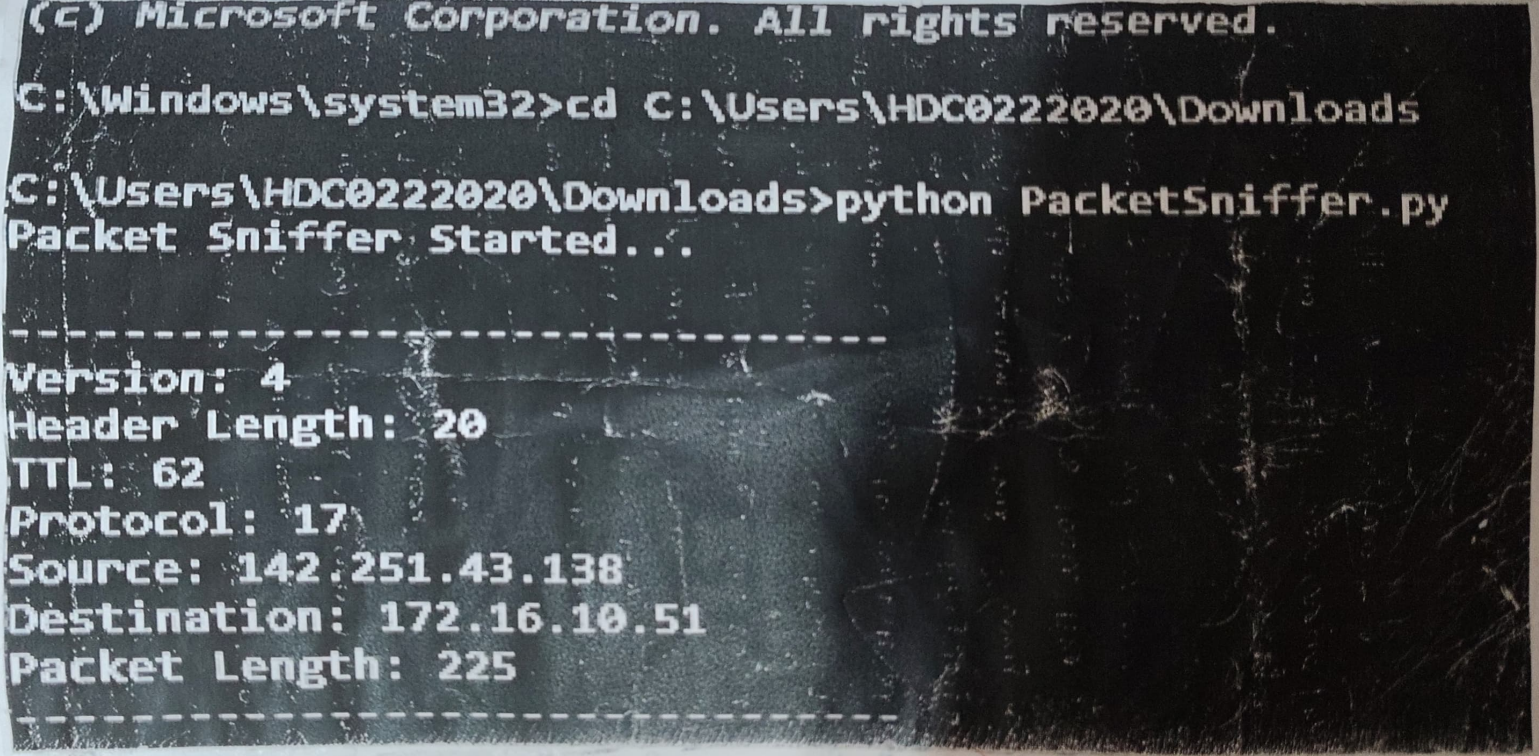
AIM:

To implement a packet sniffing using raw sockets in python that captures live network traffic and parses Ethernet, IP and transport layer (TCP/UDP/ICMP) headers

THEORY:

Raw socket: Normal sockets (SOCK_STREAM, SOCK_DGRAM) let an application send/rece data through the OS's protocol stack (TCP/UD building lower-layer headers. A raw socket (SOCK_RAW) bypass this abstraction - it giv programs direct access to packets at the IP or even the ethernet (datalink) layer including all headers. This is why raw sockets are the basis of tools like wireshark, tcpdump and custom sniffe

Why root/admin privileges ar required: raw sockets can read/inje arbitary traffic on the network.


(c) Microsoft Corporation. All rights reserved.
C:\Windows\system32>cd C:\Users\HDC0222020\Downloads
C:\Users\HDC0222020\Downloads>python PacketSniffer.py
Packet Sniffer Started...
Version: 4
Header Length: 20
TTL: 62
Protocol: 17
Source: 142.251.43.138
Destination: 172.16.10.51
Packet Length: 225

RESULT :

Packet Sniffering using raw Sockets in Python is completed and verified